Research Plan: Lamb Waves in a Magneto-Electro-Elastic Plate

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Problem statement

Consider a thin plate of thickness 2h made of a magneto-electro-elastic (MEE) composite, for example a mixture of a piezoelectric phase and a piezomagnetic phase. Both faces of the plate are free. The material is elastic, piezoelectric, piezomagnetic and thermally conducting, and the electric and magnetic fields are coupled through the strain. Heat conduction follows the Moore–Gibson–Thompson (MGT) model, and size effects are described by the nonlocal strain gradient theory, which has two length scales.

Study the Lamb waves that travel along this plate. For the symmetric and antisymmetric modes, find the phase velocity and the attenuation as functions of frequency and plate thickness. Show how they change with the volume fraction of the piezomagnetic phase, the nonlocal length, the gradient length and the thermal relaxation time, and compare different electric and magnetic face conditions.

Understanding the problem

This section explains every term used in the problem statement, one by one. If a word in a paper looks unfamiliar later, come back here first.

Why a plate? Many wave papers treat a half-space, a block of material that goes on forever below its surface. Real devices use thin plates or films, and a plate behaves differently because the wave keeps bouncing between its two faces. Why a magneto-electro-elastic material? It links three things that normally do not talk to each other, deformation, electricity and magnetism, so a wave in it carries all three at once.

Wave and plate

A wave is a disturbance that moves through a material and carries energy with it. Shake one end of a long rope and the bump runs along the rope, while the rope itself stays where it is. A plate is a flat body whose thickness is much smaller than its length and width, like a steel ruler or a sheet of tin.

Lamb wave

In a plate, a wave keeps reflecting from the top and bottom faces. The reflected parts add up and form a pattern that moves along the plate. This guided wave is called a Lamb wave, after Horace Lamb, who described it in 1917. The ringing you hear when you tap a thin metal sheet is the whole sheet vibrating, not only its surface.

Frequency, wavenumber and speed

The angular frequency ω is 2π times the number of vibrations per second. The wavenumber k is 2π divided by the wavelength. The phase velocity is $c = \omega/k$, which is the speed at which a crest moves along the plate.

Dispersion

In a plate, waves of different frequencies travel at different speeds. This is called dispersion. A glass prism shows the same idea with light: each colour moves at a slightly different speed in glass, so white light spreads into a rainbow. A dispersion curve is a graph of frequency against wavenumber, and most of our figures are built from it.

Modes

At a given frequency, a plate can carry a wave in more than one pattern, and each pattern is called a mode. In the usual case, symmetric modes stretch and squeeze the plate along its length, and antisymmetric modes bend it like a flapping sheet. They are named S0, S1, S2 and so on, and A0, A1, A2 and so on. In a magneto-electro-elastic plate, whether the modes split cleanly into these two families depends on the poling direction. We will fix that direction in step 2.

Cutoff frequency

Higher modes do not exist at every frequency. Each one appears only above its own cutoff frequency. A road is a rough picture: a narrow road carries one line of vehicles and a wider road carries several. At low frequency the plate carries only the first modes, and more become possible as the frequency rises.

Piezoelectricity

Some crystals produce a voltage when they are squeezed, and they change shape when a voltage is applied to them. The spark in a gas lighter comes from pressing such a crystal, and a quartz watch uses the opposite effect. Barium titanate ($BaTiO_3$) is a common

piezoelectric material in this kind of composite.

Piezomagnetism

Some materials become magnetised when they are squeezed, and they change shape when a magnetic field is applied. The hum of a transformer is a similar effect: its iron core changes shape very slightly as the magnetic field goes back and forth. Cobalt ferrite ($CoFe_2O_4$) is a common piezomagnetic material in composites.

Magneto-electro-elastic (MEE) composite

A magneto-electro-elastic composite joins a piezoelectric phase and a piezomagnetic phase, for example $BaTiO_3$ and $CoFe_2O_4$. Neither phase alone turns a magnetic field into a voltage. In the composite it happens in two stages: a magnetic field strains the magnetic phase, that strain passes into the piezoelectric phase, and the piezoelectric phase produces a voltage. It is like a relay race, where neither runner completes the lap alone but together they do. Such composites are studied for magnetic field sensors and energy harvesters. How strong the effect is depends on how much of each phase is present, which is why the volume fraction appears in the problem statement.

Because elastic waves are far slower than light, we use the usual quasi-static approximation. The electric and magnetic fields are then described by an electric potential φ and a magnetic potential ψ.

Heat and the MGT model

Stretching a material slightly changes its temperature, and a change in temperature makes it expand or shrink. Stretch a rubber band quickly and touch it to your lip, and it feels warm. So elastic waves and heat are linked, and that is why the problem is called thermoelastic. In these composites a temperature change can also produce an electric and a magnetic response.

The classical law of heat conduction, Fourier's law, says that a temperature change is felt everywhere at once. That cannot be exactly true. Put a pan on a stove and the handle takes a while to warm up. The Moore–Gibson–Thompson (MGT) model fixes this by letting heat travel at a finite speed, using a thermal relaxation time.

Size effect

Classical elasticity says that the stress at a point depends only on the strain at that same point. That works very well for everyday objects. At micro and nano scales, tests on tiny beams and plates show behaviour that this rule cannot explain.

Nonlocal theory, started by Eringen, says the stress at a point depends on the strain in a neighbourhood around it. Strain gradient theory says it also depends on how quickly the strain changes from place to place. The nonlocal strain gradient theory uses both ideas and has two length scales, one for each. When the plate is much thicker than these lengths, the classical answer comes back. Think of a person in a crowded lift: what they feel depends on the people around them, not only on the spot where they stand.

Attenuation and complex wavenumber

If a wave loses energy as it travels, the wavenumber is no longer a real number. It is written k = k_r + i k_i. The real part k_r gives the wavelength and the speed. The imaginary part k_i tells how fast the amplitude falls, in proportion to e^(−k_i x). A struck bell is the usual picture: the sound is loud at first and then slowly fades. In our plate, heat flow is the main cause of fading. When the figures mention complex branches, they mean curves where k has both a real and an imaginary part.

The model we will use

The model is kept small so that each part can be checked on its own.

The plate has thickness 2h, and the wave travels along its length. We work in plane strain, which means the motion takes place in the length and thickness directions only. The material is a transversely isotropic MEE composite, with the poling direction taken along the thickness as in most published plate papers; we confirm this in step 2 while reading them. The top and bottom faces carry no stress. They can be thermally insulated or held at a fixed temperature, and electrically and magnetically open or short-circuited.

Inside the plate we use the elastic equations with piezoelectric and piezomagnetic coupling, the electric and magnetic equations in their quasi-static form, and the MGT heat equation with its electric and magnetic temperature coupling. The size effect enters through the nonlocal strain gradient law with two length scales, first applied to the elastic stresses.

The composite is described by effective properties for a chosen volume fraction of the piezomagnetic phase. We take these from published tables for $BaTiO_3$ and $CoFe_2O_4$ composites and do not derive them ourselves.

We leave out surface elasticity, flexomagnetoelectric effects, imperfect interfaces, layered or graded plates, memory-dependent derivatives, nonlinear effects and full 3D geometry. Each of these can be a separate paper later.

Two points must be written clearly in the paper. First, the size-effect theory asks for extra boundary conditions on the faces. We apply the usual ones, say so openly, and show how much the results change when the length scales change. Second, effective constants depend on the homogenisation method and its source. We name the source, use a range of volume fractions, and show the sensitivity.

Work plan

The work is done in six steps. After each step there is a check against a known result. Please send me that check plot before you start the next step, so that if something is wrong we know which step caused it.

Step	What you do	The check before moving on	What you gain
1. Plain elastic plate	Derive and solve the classical Rayleigh–Lamb problem	Known cutoff frequencies and thin-plate limits match; for a very thick plate the speed approaches the Rayleigh half-space speed	Confidence with the method
2. Add piezoelectricity	Include the electric potential, for open and short-circuited faces	Switching the coupling off returns step 1; the open/short speed difference gives the coupling coefficient K^2	A first result worth showing

Step	What you do	The check before moving on	What you gain
3. Add piezomagnetism	Include the magnetic potential and the magnetoelectric link, and sweep the volume fraction	Switching the magnetic coupling off returns step 2; the dispersion curves agree with a published magneto-electro-elastic plate result from the reading list	The MEE part of the paper
4. Add heat (MGT)	Include the temperature field and the MGT heat equation	Switching thermal coupling off returns step 3; the classical thermoelastic limit agrees with published results	A thermal effect on speed and fading
5. Add the size effect	Include the nonlocal strain gradient terms	Setting both length scales to zero returns step 4; nonlocal-only and gradient-only cases are checked separately	The main new contribution
6. Analysis	Track complex branches, fading, maps and sensitivity	Branches vary smoothly and root counts stay consistent along each branch	The figures that carry the paper

After step 4 we already have a verified thermal MEE plate, which is a usable base for the paper. Step 5 adds the new part. If step 5 turns out to be heavy, we will decide together what to do.

Calculations and figures

For each wavenumber we find the complex frequencies at which the plate can carry a wave, and then follow each branch from one wavenumber to the next so that curves do not get mixed up. I will give you working code for this part, so your time goes into

understanding the results and making the figures.

Figure	What it shows	Why the paper needs it
Dispersion curves	Frequency against wavenumber for each mode	The base result
Complex branches	Real and imaginary parts together	Shows the full wave behaviour, not only the real curves
Phase speed against frequency–thickness	Where each mode switches on and how it settles	Shows true plate behaviour
Volume fraction maps	How the share of the piezomagnetic phase changes speed and coupling	The MEE result of the paper
Fading against frequency	Effect of heat on how quickly waves die out	Links to device performance
Thickness–frequency–speed surface	The whole parameter landscape in one view	Rich information in one figure
Length-scale contour maps	Separate effects of the two size parameters	The main result of the size model
Coupling coefficients against thickness	Electric and magnetic open versus short-circuit differences	Useful for sensor design
Mode shapes	Displacement, electric potential, magnetic potential and temperature across the thickness	Builds physical understanding
Sensitivity ranking	Which constant changes the speed most	Shows how far results can be trusted
Verification table	Each limiting case and its match	Builds reviewers' trust

All figures use one colour scheme, one font size and a resolution of 300 dpi, so the paper looks like one consistent piece of work.

Possible difficulties

These are the problems I expect along the way. None of them is unusual.

Difficulty	What it looks like	How we handle it
Closely related papers already exist	Lamb waves in magneto-electro-elastic plates, size-dependent MEE plates and thermal MEE plates have each been published separately	Read these papers fully first, state exactly which combination is new, and do not claim more
Extra boundary conditions in the size-effect theory	A reviewer may ask why only the usual conditions are applied	We state our choice openly and show how little or how much the results move when the length scales change
Effective constants differ between sources	Results change with the homogenisation method	Name the source, use a range of volume fractions, and show the sensitivity
Electric and magnetic face conditions	Open and short-circuit choices give different curves, and it is easy to mix them up	Keep a small table of the chosen conditions for every figure
A missing or tangled branch	A curve jumps or two curves swap places	Follow each branch step by step and check the root count along the way
Too much at once	The full model can feel large	Work step by step, and treat step 5 as the last addition

If you get stuck at any step, send me the plot that looks wrong and we will go through it together.

Reading list

Nine items are enough to start. Read the first two first, since they show how a magneto-electro-elastic plate is set up and solved, then item 5 for how the nonlocal strain gradient idea is used for these materials. Read for the idea first and the algebra afterwards.

#	Paper	Why read it
1	Chen, Pan and Chen, Wave propagation in magneto-electro-elastic multilayered plates, International Journal of Solids and Structures, 2007	The classical dispersion relation for MEE plates with traction-free faces; a source for checking step 3
2	Zhang, Shen and Yu, Dispersion characteristics of Lamb wave in a multilayered magnetoelectroelastic plate with an imperfect interface, Journal of Intelligent Material Systems and Structures, 2025	A recent Lamb wave study on $BaTiO_3/CoFe_2O_4$ plates, with coupling factors and boundary conditions
3	Zakharenko, On discovery of extra four new dispersive SH-waves in magnetoelectroelastic plates, arXiv, 2018	Shows the kinds of wave modes that appear in MEE plates, and how the face conditions are set
4	Wu, Zhang, Chen and Zhang, Surface effects on anti-plane shear waves propagating in magneto-electro-elastic nano-plates, arXiv, 2021	A size-dependent MEE plate paper using a surface-based model; compare its approach with ours
5	Critical Temperatures for Vibrations and Buckling of Magneto-Electro-Elastic Nonlocal Strain Gradient Plates, Nanomaterials, 2021	How the nonlocal strain gradient law is written for MEE plates, and which temperature coupling is used
6	Ray, Dispersion curves for Rayleigh–Lamb waves in a micro-plate considering strain gradient elasticity, Wave Motion 86, 2019	How a strain gradient plate is set up and how its dispersion is built
7	Lim, Zhang and Reddy, A higher-order nonlocal elasticity and strain gradient theory and its applications in wave propagation, J. Mech. Phys. Solids 78, 2015	The source of the nonlocal strain gradient theory; confirm the exact citation before citing
8	Shaat, Infeasibility of the nonlocal strain gradient theory for applied Physics, arXiv, 2017	A critique of the theory; good to know so we can answer it if a reviewer raises it

#	Paper	Why read it
9	Our own MGT papers, for example Gupta et al., Piezo-semiconductor-void interplay in the size-dependent dynamics of Rayleigh waves within the Moore–Gibson–Thompson framework, IJNMHFF 35(9), 2025	The MGT heat equation and its coupling that we will reuse

Outline of the paper

1. **Introduction:** why plates matter, what magneto-electro-elastic materials add, what size-dependent theories add beyond the classical Eringen model, and what the closest earlier papers did and did not cover.
2. **Formulation:** the plate, the field equations for elasticity, the electric and magnetic potentials and MGT heat, the size-effect law, and the boundary conditions with the clear note on extra conditions.
3. **Dispersion relation:** the determinant, and how the modes split into families, depending on the poling direction.
4. **Numerical method and verification:** how roots are found and followed, and the table of limiting cases.
5. **Results:** dispersion and complex branches, volume fraction maps, fading, the thickness surface, length-scale maps, coupling coefficients, mode shapes and sensitivity.
6. **Conclusions:** what each claim showed, the limitations, and future work such as surface effects, imperfect interfaces and graded plates.
7. **Supplementary material:** the code and the full parameter table, so anyone can reproduce the results.

The claims to keep in view while working, to be finalised after the reading of items 1 to 5 above:

- A coupled magneto-electro-elastic plate model with MGT heat and a two-length-scale size effect.
- Real and complex branches followed together, with fading and mode behaviour.
- Every step verified against a known limit.